

Group Project
SECP 3223 Data Analytics Programming
20242025 – Sem 1
Due Date: 21st January 2025

DATA THEME: CLIMATE CHANGE

Data Theme: Healthcare Access and Outcomes

Healthcare is a critical issue for every nation, with access to quality healthcare often determining overall well-being and economic productivity. Globally, healthcare systems face a range of challenges including unequal access, shortages of medical professionals, funding issues, and disparities in healthcare outcomes. This project should examine data related to healthcare availability, healthcare costs, public health metrics, and patient outcomes to uncover hidden patterns and predict potential future challenges.

You will be working on a data analytics project using datasets that can be downloaded from the World Health Organization Data website:

(https://cdn.who.int/media/docs/default-source/gho-documents/world-health-statistic-reports/2024/web_download.xlsx?sfvrsn=4e7bbebd_1). The following are the instructions:

1. This group project has a maximum of 5 students per group.
2. This project aims to expose students to the data analytics process starting from data preparation, data wrangling, visualization and machine learning application.
3. This project is divided into several tasks:
 - a. Find 1 or more suitable dataset(s) which contains enough data to practice data preparation and analysis (at least 1000 rows)
 - b. Formulate research questions(s) from the dataset. What do you want to analyze?
 - c. Do appropriate data cleaning, preparation and wrangling.
 - d. Do appropriate data aggregation and group operations.
 - e. Visualize your analysis using appropriate visualization.

f. Perform appropriate machine learning method to answer your research question in (b).
*You are welcome to explore any methods/functions/techniques out of this course syllabus.

4. The submission of the project will be via e-learning. Documents to submit:
 - a. Python codes and report in Jupyter Notebook (ipynb format) which explain in detail all the tasks in step 3.
 - b. Initial dataset (before performing any operations)
 - c. Video presentation (any software of your choice) – the duration is **max 10 minutes**.
 - Please zip items a – b and upload in e-learning
 - Item c to be uploaded in your Google Drive and the link will be shared in your Jupyter Notebook (item a). Please make sure that the video can be viewed by the lecturer.
5. Please refer to the rubric given to see the assessment method of this project.

Please make sure that all group members' details (name and matric number) are written in all submitted documents (Jupyter Notebook file and video presentation).

MARK SHEET FOR GROUP PROJECT

ITEMS	MARKS		
	1 - 3	4 - 7	8 - 10
<i>PART A – REPORT</i>			
<i>Section i</i>			
Dataset			
Research Question			
Section Total	(/ 20 × 5) =		
<i>Section ii</i>			
Data Cleaning and Preparation			
Data Aggregation and Group Operations			
Analysis and visualization			
Machine Learning			
Section Total	(/ 40 × 25) =		
<i>Section iii</i>			
Conclusion			
Overall Writing			
Section Total	(/ 20 × 5) =		
TOTAL (PART A)			
<i>PART B – ipynb PYTHON CODE</i>			
Code Review			
TOTAL (PART B)	(/ 10 × 5) =		
<i>PART C – VIDEO PRESENTATION</i>			
Verbal Presentation			
Visual Presentation			
Overall Presentation			
TOTAL (PART C)	(/ 30 × 10) =		
OVERALL TOTAL	(/ 50 × 25) =		

RUBRIC FOR GROUP PROJECT

ITEMS	MARKS		
	1 - 3	4 - 7	8 - 10
<i>PART A - REPORT</i>			
Dataset	Too simple with not enough data to perform data preparation and analysis	Appropriate with enough data to perform data preparation and analysis	Good choice of dataset with large amount of data which can be used to perform complete data preparation and analysis
Research Question	Questions overly simplistic, unrelated, or unmotivated	Questions appropriate, coherent, and motivated	Questions well motivated, interesting, insightful, and novel
Data Cleaning and Preparation	Overly simplistic or incomplete	Appropriate and complete	Appropriate, complete and advanced
Data Aggregation and Group Operations	Overly simplistic or incomplete	Appropriate and complete	Appropriate, complete and advanced
Analysis and visualization	Choice of analysis is overly simplistic or incomplete. Inappropriate choice of plots; poorly labelled plots; plots missing	Analysis appropriate. Plots convey information but lack context for interpretation	Analysis appropriate, complete, advanced, and informative. Plots convey information correctly with adequate and appropriate reference information
Machine Learning	Overly simplistic or incomplete	Appropriate and complete	Appropriate, complete and advanced
Conclusion	Conclusions are missing, incorrect, or not based on analysis	Conclusions relevant, but partially correct or partially complete	Relevant conclusions explicitly tied to analysis and to context
Overall Writing	Explanation is illogical, incorrect, or incoherent	Explanation is correct, complete, and convincing	Explanation is correct, complete, convincing, and elegant
<i>PART B – ipynb PYTHON CODE</i>			
Code Review	Code is messy and poorly organized; unused or irrelevant code distracts when reading code. Variables and functions names do not help to understand code.	Code is reasonably well organized. There is little unused or irrelevant code, or this code has been moved out of the main project files. Variable and function names	Code very well organized. No irrelevant or distracting code. Variable and function names have clear relationship to their purpose in the

		generally meaningful and helpful for understanding.	code. Code is easy to read and understand.
<i>PART C – VIDEO PRESENTATION</i>			
Verbal Presentation	Illogical, incorrect, or incoherent.	Partially correct but incomplete or unconvincing	Correct, complete, and convincing
Visual Presentation	Cluttered, disjoint, or illegible	Readable and clear	Appealing, informative, and crisp
Overall Presentation	Verbal and visual presentation unrelated	Verbal and visual presentation related	Verbal and visual presentation clearly related